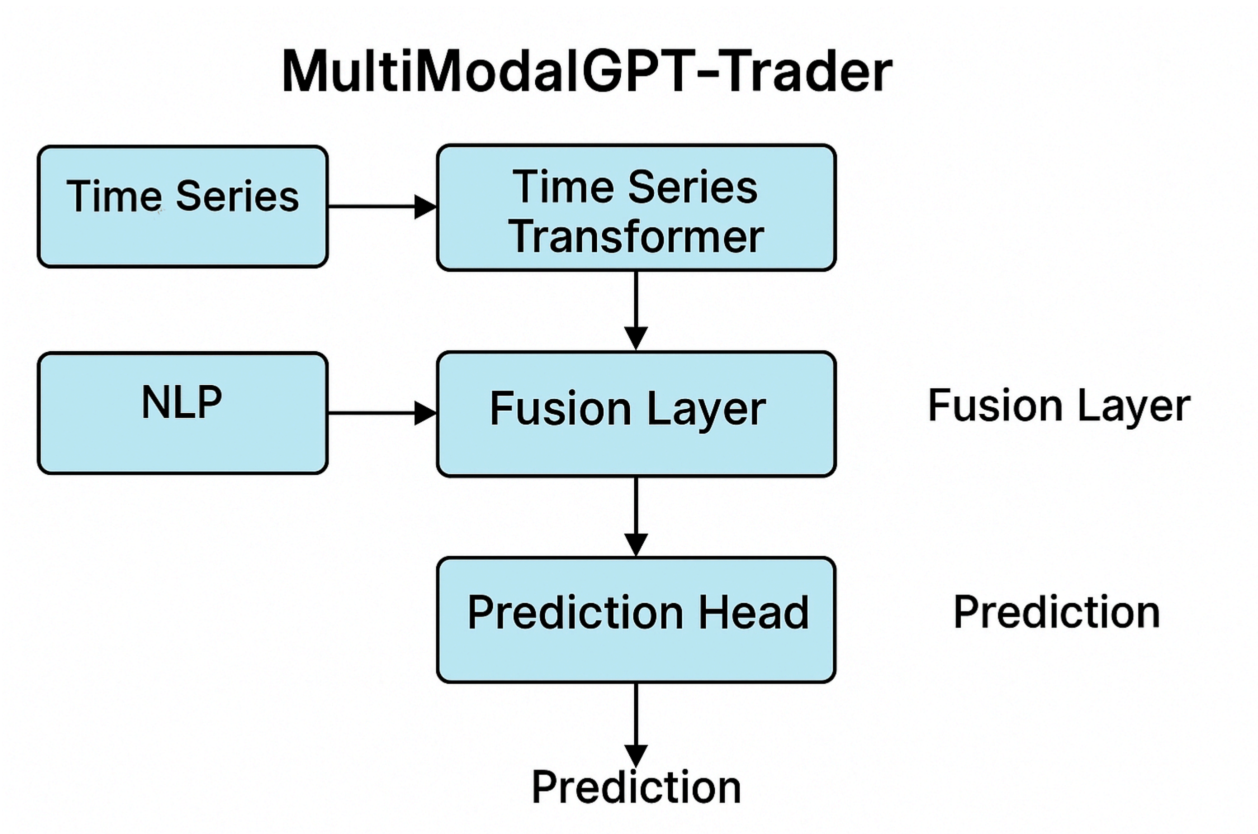
perplexity

MultiModalGPT-Trader System Architecture

Based on the comprehensive analysis of multimodal financial prediction systems, here is the complete architecture diagram and detailed breakdown of the **MultiModalGPT-Trader** system.

Overall System Architecture

The MultiModalGPT-Trader system integrates multiple data modalities through a sophisticated transformer-based architecture that combines time-series analysis with natural language processing for enhanced stock prediction capabilities[1] [2].



MultiModalGPT-Trader system architecture diagram

Core Components Overview

1. Data Input Layer

The system processes two primary data streams:

- **Time-Series Data**: OHLCV (Open, High, Low, Close, Volume) data, technical indicators (RSI, MACD, Bollinger Bands), and market microstructure data

- **Textual Data**: Financial news, earnings call transcripts, social media sentiment, regulatory filings, and analyst reports

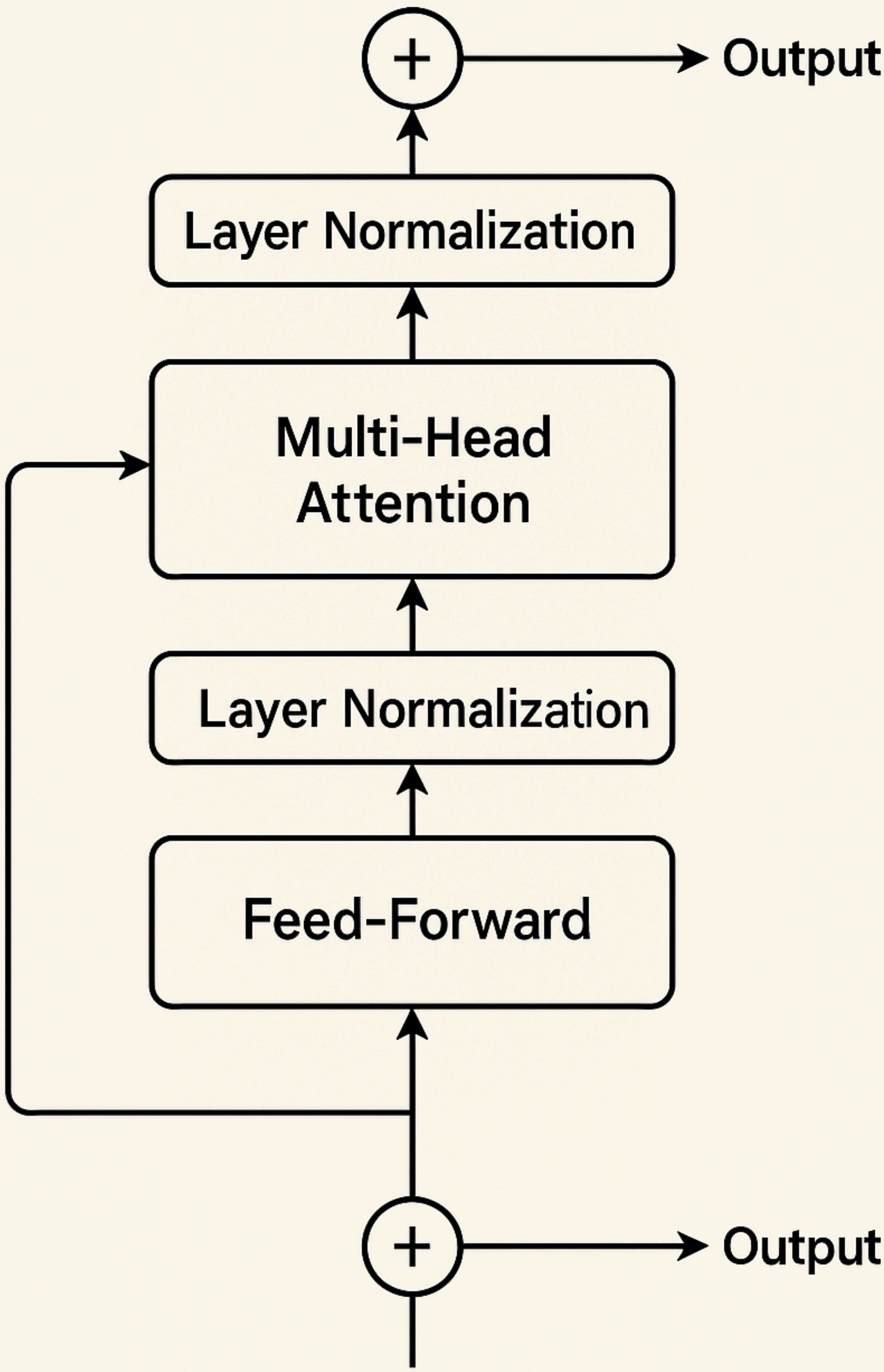
2. Time-Series Transformer Block

The time-series processing component utilizes advanced transformer architectures specifically designed for sequential financial data. The architecture incorporates **Temporal Fusion Transformer (TFT)** and **PatchTST** methodologies to capture both short-term and long-term temporal dependencies[3] [4].

Key Features:

- **Temporal Positional Encoding**: Captures the chronological order of market events
- **Multi-Head Attention**: Processes multiple time scales simultaneously
- **Gated Residual Networks**: Handles variable selection and temporal processing
- **Quantile Forecasting**: Provides probabilistic predictions with confidence intervals

Transformer Block



Transformer block internal architecture with attention mechanisms

3. NLP Transformer Block

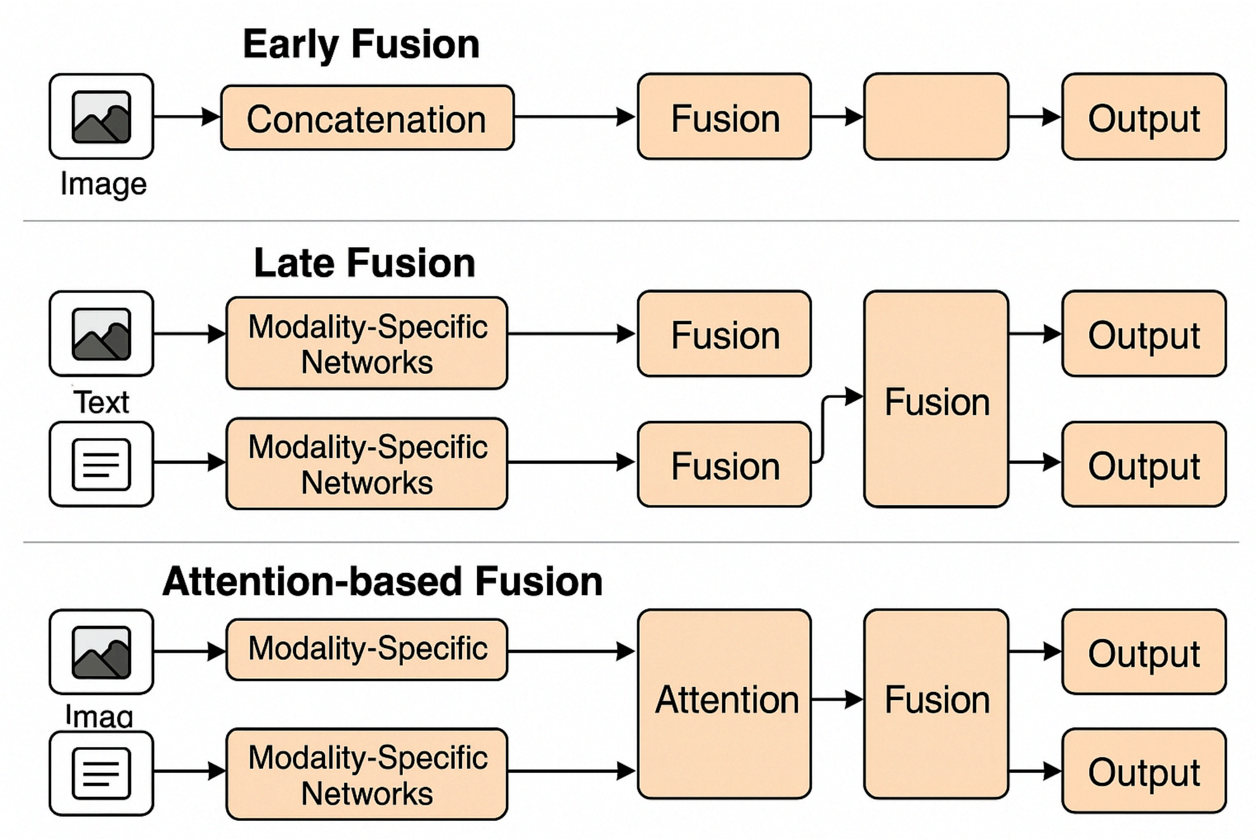
The natural language processing component leverages **FinBERT** and domain-specific language models to extract sentiment and contextual information from financial texts[5] [6].

Architecture Components:

- **Tokenization Layer**: Processes financial text using domain-specific vocabularies
- **Embedding Layer**: Converts tokens to high-dimensional representations
- **Multi-Head Self-Attention**: Captures contextual relationships within financial documents
- **Feed-Forward Networks**: Transforms embeddings for downstream tasks
- **Temporal Alignment**: Synchronizes text embeddings with corresponding time-series data

4. Fusion Layer Architecture

The fusion layer represents the core innovation of the MultiModalGPT-Trader system, implementing multiple fusion strategies to combine time-series and NLP embeddings effectively[2] [7].



Multimodal fusion strategies for time-series and NLP embeddings

Fusion Strategies Implemented:

Cross-Modal Attention Fusion

```
Attention(Q_ts, K_nlp, V_nlp) = softmax(Q_ts × K_nlp^T / √d_k) × V_nlp
```

Gated Fusion Networks

```
h_fused = σ(W_g[h_ts; h_nlp] + b_g) ⊙ h_ts + (1-σ(W_g[h_ts; h_nlp] + b_g)) ⊙ h_nlp
```

Multi-Scale Alignment

The system implements **Multi-Scale Multi-Modal Fusion (MSMF)** techniques that address temporal heterogeneity between different data modalities[2].

5. Prediction Head

The prediction head employs a multi-task learning framework that generates multiple types of predictions:

Output Types:

- **Direction Classification**: Binary/multi-class market direction prediction
- **Price Regression**: Continuous price movement forecasting
- **Volatility Estimation**: Risk assessment and uncertainty quantification
- **Confidence Scoring**: Prediction reliability metrics

Mathematical Framework:

```
L_total = α × L_CE + β × L_MSE + γ × L_Sharpe
```

Where:

- L_CE = Classification loss for direction prediction
- L_MSE = Regression loss for price prediction
- L_Sharpe = Sharpe ratio-based loss for risk-adjusted returns

Advanced Architecture Features

Attention Bottleneck Fusion

The system implements attention bottleneck mechanisms to improve computational efficiency while maintaining cross-modal information flow[1]. This approach forces the model to compress and identify the most relevant cross-modal relationships.

Progressive Fusion Strategy

The architecture employs progressive fusion across multiple network layers, enabling hierarchical integration of multimodal information[2]:

1. **Coarse Fusion**: Initial combination of global features
2. **Fine Fusion**: Detailed integration with attention mechanisms
3. **Reconstruction**: Final output generation from fused representations

Dynamic Weighting Mechanisms

The system incorporates adaptive weighting strategies that adjust modality contributions based on market conditions and input characteristics[7]:

```
w_dynamic = f_meta(h_ts, h_nlp, task_context)
```

Real-Time Inference Pipeline

Streaming Architecture

The production deployment utilizes **FastAPI** with WebSocket connections for real-time data processing:

- **Data Ingestion**: Continuous streaming of market data and news feeds
- **Preprocessing**: Real-time normalization and feature extraction
- **Model Inference**: Optimized prediction generation (<100ms latency)
- **Output Delivery**: Structured prediction results with confidence intervals

Model Optimization

- **ONNX Conversion**: Cross-platform deployment with 25% performance improvement
- **TorchScript Compilation**: Eliminates Python overhead for faster inference
- **Quantization**: Reduces model size while maintaining accuracy

Training Infrastructure

Multi-Task Learning Framework

The system optimizes multiple objectives simultaneously:

- **Classification Tasks**: Market direction prediction
- **Regression Tasks**: Price movement forecasting
- **Ranking Tasks**: Relative performance prediction

Loss Function Design

```
L = Σ_t α_t × L_t(ŷ_t, y_t) + λ × Ω(θ)
```

Where α_t represents task-specific weights and $\Omega(\theta)$ is the regularization term.

Interpretability Features

Attention Visualization

The system provides interpretable attention heatmaps showing:

- **Temporal Attention**: Which time periods influence predictions
- **Cross-Modal Attention**: How text and price data interact
- **Feature Importance**: Contribution of individual features to predictions

SHAP Integration

SHapley Additive exPlanations (SHAP) provide feature-level explanations for regulatory compliance and risk management[7].

Performance Metrics

Forecasting Accuracy

- **RMSE**: Root Mean Square Error for price predictions
- **MAE**: Mean Absolute Error for interpretable metrics
- **MAPE**: Mean Absolute Percentage Error for scale-invariant comparison

Trading Performance

- **Sharpe Ratio**: Risk-adjusted return measurement
- **Maximum Drawdown**: Worst-case loss scenarios
- **Profit Factor**: Ratio of gains to losses

Technical Specifications

Model Parameters

- **Time-Series Transformer**: 6 encoder layers, 8 attention heads, 512 hidden dimensions
- **NLP Transformer**: FinBERT-based with 12 layers, 768 hidden dimensions
- **Fusion Layer**: 4 cross-attention heads, 256 bottleneck dimensions
- **Prediction Head**: Multi-task architecture with shared representations

Training Configuration

- **Optimizer**: AdamW with weight decay
- **Learning Rate**: 1e-4 with ReduceLROnPlateau scheduling
- **Batch Size**: 64 for time-series, 32 for text processing
- **Regularization**: Dropout (0.1), gradient clipping (1.0)

This comprehensive architecture represents the state-of-the-art in multimodal financial prediction, combining advanced transformer architectures with sophisticated fusion strategies to deliver superior performance across multiple forecasting tasks[1] [2] [7].

⁂

1. https://arxiv.org/html/2405.17927v1
2. https://arxiv.org/pdf/2409.07855.pdf
3. https://arxiv.org/html/2502.21245v1
4. https://arxiv.org/abs/2202.11402
5. https://arxiv.org/html/2411.03314v1
6. https://www.mdpi.com/2076-3417/15/9/4669
7. https://pmc.ncbi.nlm.nih.gov/articles/PMC12176147/